

Program: Diploma in Artificial Intelligence & Machine Learning	
Course Code: 6342B	Course Title: Fundamentals of Soft Computing
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To become familiar with neural networks and understand its features and applications.
- To get exposed to the mathematical background for solving problems using neural network learning algorithms.
- To get exposed to the ideas of fuzzy logic and also provide knowledge about the use of inference systems and approximate reasoning in solving complex tasks.
- To familiarize with genetic algorithms useful for solving complex optimization.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Mathematics		Mathematics I & II	1,2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize fundamental concepts of soft computing	7	Understanding
CO2	Explain the concepts of Artificial Neural Network (ANN)	18	Understanding
CO3	Demonstrate the concepts of fuzzy sets	18	Understanding
CO4	Concepts of genetic algorithms	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Descriptions	Duration (Hours)	Cognitive Level
------------------------	---------------------	-------------------------	------------------------

CO1	Summarize fundamental concepts of soft computing		
M1.01	Understand the concept of soft computing.	2	Understanding
M1.02	Differentiate "Soft" computing and "Hard" computing	1	Understanding
M1.03	Discuss conventional AI to Computational Intelligence.	2	Understanding
M1.04	Applications of Soft computing techniques	2	Understanding
Contents: Basic concepts of soft computing -Definition, Elements of soft computing, Characteristics of Soft Computing. Differentiate Soft Computing and Hard Computing, Conventional AI, Computational Intelligence, Applications of Soft computing.			
CO2	Outline concepts of Artificial Neural Network (ANN)		
M2.01	Understand Artificial Neural Networks.	3	Understanding
M2.02	Outline the basic models of ANN.	5	Understanding
M2.03	Summarize important terminologies of ANN	6	Understanding
M2.04	Understand basic networks in Supervised Learning	4	Understanding
Contents: Artificial Neural Networks-Familiarization, Advantages, application and scope, Basic Models of ANN -connections- learning- activation functions (identity, binary step, bipolar step, sigmoidal, ramp). Important terminologies of ANN – weights, bias, threshold, learning rate, momentum factor, vigilance parameter, notations-McCulloch-Pitts Neuron Overview of Supervised learning network- Perceptron Network- Adaline - Madaline			
CO3	Demonstrate fuzzy logic and sets		
M3.01	Familiarise fuzzy sets	2	Understanding
M3.02	Illustrate fuzzy Operations and Properties	5	Understanding
M3.03	Summarize Fuzzy Relations and Membership functions	6	Understanding
M3.04	Understand Fuzzy Rules and Fuzzy logic	4	Understanding
Contents: Fuzzy Sets: Operations and Properties, Fuzzy Relations-Cardinality, operations on fuzzy relations. Membership functions- Features of the membership functions			

Fuzzy Rules and Fuzzy Reasoning Fuzzy Inference Systems, Fuzzy Logic.			
CO4	Understand concepts of Genetic Algorithms		
M4.01	Define Genetic algorithm	1	Understanding
M4.02	Understand Biological terminologies	2	Understanding
M4.03	Understand basic terminologies and working of genetic algorithms..	3	Understanding
M4.04	Explain the Simple Genetic algorithm	3	Understanding
M4.05	Illustrate Operators in Genetic Algorithm	4	Understanding
M4.06	Outline Applications of Genetic Algorithm	1	Understanding
Contents: Genetic Algorithms(GA)-Basic biological terminologies, Terminologies and working of genetic algorithms.Simple genetic algorithm, Operators in GA– encoding - selection- recombination - mutation. Applications of Genetic Algorithms.			

Text / Reference

T/R	Book Title/Author
T1	S.N. Sivanandam, S.N. Deepa, “ Principles of Soft Computing”, Wiley India, 2nd Ed.,2011.
T2	Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani, ”Neuro-Fuzzy and Soft Computing: A Computational Approach to Learning and Machine Intelligence, Prentice Hall, Upper Saddle River, NJ, ©1997.
T3	Mitchell Melanie, “An Introduction to Genetic Algorithms”, MIT Press, 1998.
T4	B. Yegnanarayana, “Artificial Neural Networks”, PHI, 1999.
R1	Laurene Fauseett: ”Fundamentals of Neural Networks”, Prentice Hall India, New Delhi,1994.
R2	David A. Coley, “Introduction to Genetic Algorithms for Scientists and Engineers”, World Scientific Publishers, 1999

Online Resources

Sl.No	Website Link
1	https://www.elprocus.com/soft-computing/
2	https://www.tutorialspoint.com/adaline-and-madaline-network
3	https://www.turing.com/kb/genetic-algorithm-applications-in-ml